Pointers and Arrays

```
#include<stdio.h>
void main()
{
int num[ ] = { 24, 34, 12, 44, 56, 17 } ;
int i ;
for ( i = 0 ; i <= 5 ; i++ )
{
printf ( "\nelement no. %d ", num[i] ) ;
printf ( "address = %p", &num[i] ) ;
}
}
```

```
element no. 24 address = 3589325760
element no. 34 address = 3589325764
element no. 12 address = 3589325768
element no. 44 address = 3589325772
element no. 56 address = 3589325776
element no. 17 address = 3589325780
```

```
#include<stdio.h>
void main()
{
int num[ ] = { 24, 34, 12, 44, 56, 17 } ;
int i ;
for ( i = 0 ; i <= 5 ; i++ )
{
printf ( "\nelement no. %d ", i[num] ) ;
printf ( "address = %u", &i[num] ) ;
}
}
```

```
element no. 24 address = 3589325760
element no. 34 address = 3589325764
element no. 12 address = 3589325768
element no. 44 address = 3589325772
```

```
element no. 56 address = 3589325776
element no. 17 address = 3589325780
```

Passing an Entire Array to a Function

```
#include<stdio.h>
void display ( int *j, int n )
{
int i ;
for (i =0;i <n;i++)
{
printf("%d ",*j);
j++ ; /* increment pointer to point to next element */
}
}
void main()
{
int num[] = { 24, 34, 12, 44, 56, 17 } ;
display(&num[0], 6 );
//display(num, 6 );
}
```

24 34 12 44 56 17

Printing the Entire String using a Pointer

```
#include <stdio.h>
int main() {
    char name[] = "Computer";
    char *ptr = name;

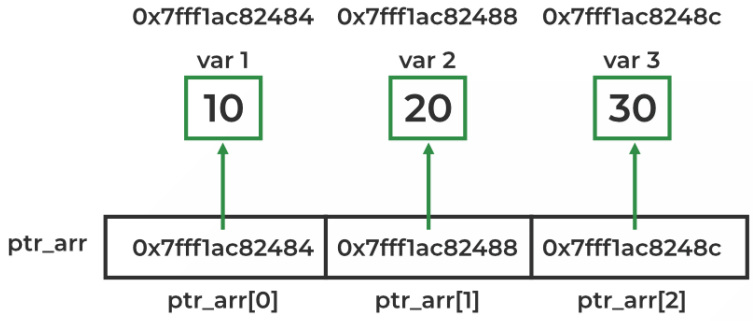
    while (*ptr != '\0') {
        printf("%c", *ptr);
        ptr++;
    }
    return 0;
```

```
}
```

Computer

Array of Pointers to Integer

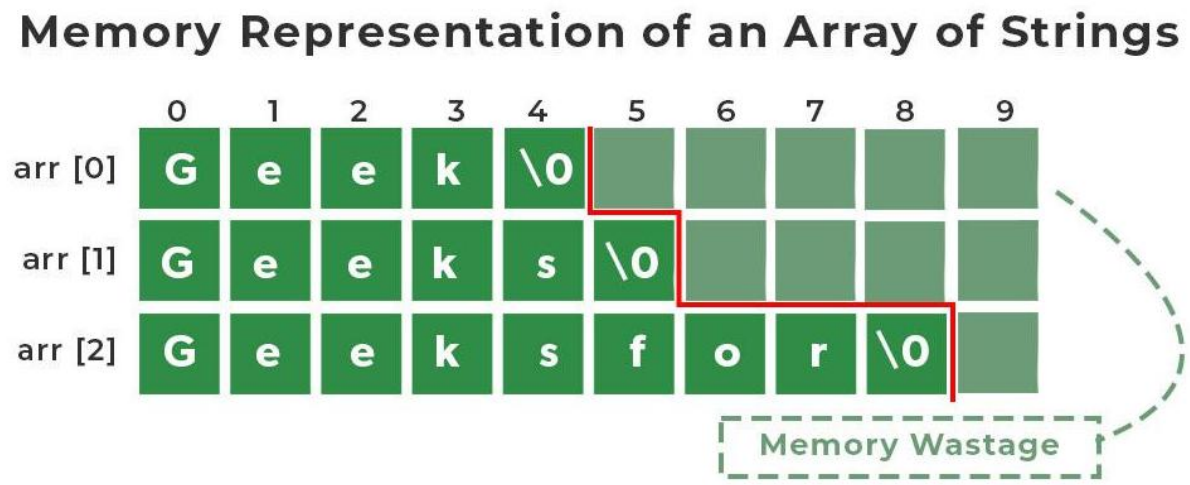
```
#include <stdio.h>
int main()
{
    int var1 = 10;
    int var2 = 20;
    int var3 = 30;
    // array of pointers to integers
    int* arr[3] = { &var1, &var2, &var3 };
    for (int i = 0; i < 3; i++) {
        printf("Value:  %d\tAddress: %p\t", *arr[i], arr[i]);
        printf("Pointer Address: %p\n", &arr[i]);
    }
    return 0;
}
Value:  10     Address: 0xffffc6d390c8     Pointer Address: 0xffffc6d390a8
Value:  20     Address: 0xffffc6d390c4     Pointer Address: 0xffffc6d390b0
Value:  30     Address: 0xffffc6d390c0     Pointer Address: 0xffffc6d390b8
```



Array of Pointers to Character

char* arr[5] = { "gfg", "geek", "Geek", "Geeks", "GeeksforGeeks" }

C Program to print Array of strings without array of pointers




```
#include <stdio.h>
int main()
{
    char str[3][10] = { "Geek", "Geeks", "Geekfor" };
    printf("String array Elements are:\n");
    for (int i = 0; i < 3; i++) {
        printf("%s\n", str[i]);
    }
    return 0;
}
```

```
String array Elements are:
Geek
Geeks
Geekfor
```

C Program to print Array of strings with array of pointers

```
#include <stdio.h>
int main() {
    char* arr[] = { "geek", "Geeks", "Geeksfor" };
    for (int i = 0; i < 3; i++) {
        printf("Value:%s\t Address:%p\n", arr[i],arr[i]);
    }
    return 0;
}
```

```
Value:geek    Address:0xca534b5608a8
Value:Geeks   Address:0xca534b5608b0
Value:Geeksfor        Address:0xca534b5608b8
```